FIREWISE-X

Flame in Freefall

AI-Powered Microgravity Fire Intelligence
& Astronaut Safety Platform

Project Blueprint | Built on NASA PSI-25 microgravity combustion experiments
Understand the Flame. Anticipate the Risk. Protect the Mission.

1. MAIN GOAL OF THE WEBSITE

FIREWISE-X is an AI-powered scientific decision-support platform designed to help astronauts, spacecraft safety engineers, and researchers understand, analyze, predict, and investigate fire behavior in microgravity environments before and during space missions.

The primary goal is to develop an intelligent system that learns from NASA's PSI-25 microgravity combustion experiments and uses computer vision, machine learning, temporal analysis, and explainable AI to provide actionable scientific insights into spacecraft fire behavior.

Unlike a conventional dataset visualization website, FIREWISE-X will operate as a functional AI intelligence platform.

The system should allow users to:

- Analyze experimental and uploaded combustion videos.
- Detect and track flame regions.
- Understand flame growth, movement, and extinction behavior.
- Evaluate combustion conditions using experimental parameters.
- Predict supported combustion outcomes.
- Investigate possible fire-related hazards.
- Receive AI-assisted explanations and safety recommendations.
- Explore hypothetical combustion scenarios before a space mission.
- Generate scientific reports for researchers and mission planners.

The platform should have two main objectives:

1. PRE-MISSION INTELLIGENCE: Help researchers and spacecraft safety engineers investigate combustion behavior and evaluate different experimental conditions before a mission.
2. IN-MISSION DECISION SUPPORT: Help astronauts and mission operators interpret available sensor measurements and visual evidence, understand potential abnormal combustion behavior, and consult AI-assisted response guidance.

> *FIREWISE-X is a research prototype, not a certified spacecraft fire suppression or emergency control system. It must not replace NASA-approved spacecraft safety procedures or human mission control decisions.*

2. HOW NASA PSI-25 WORKS AS THE BACKEND SCIENTIFIC FOUNDATION

The NASA PSI-25 dataset will remain in the backend. Users do not need to browse the original dataset or understand its folder structure.

The downloaded dataset includes:

- Engineering Documents

- Experimental Table
- Investigation Metadata Files
- Raw Data
- Reference Data
- Reports
- Science Documents

These resources have different responsibilities.

A. Raw Data

Used to obtain original experimental observations, including available videos, measurements, and other experimental files.

B. Experimental Table

Used to extract structured experimental variables and available measured outcomes.

C. Investigation Metadata

Used to identify experiments, experimental conditions, and relationships between data files.

D. Engineering Documents

Used by developers and researchers to understand instrumentation, sensor definitions, units, and technical specifications.

E. Reference Data

Used for scientific context, reference measurements, and data interpretation where applicable.

F. Science Documents

Used to understand combustion behavior and establish scientific context for model development.

G. Reports

Used as supporting scientific evidence and for validating the interpretation of experimental outcomes.

Backend data pipeline

1. Import the NASA PSI-25 files into the backend.
2. Extract and organize experimental records.
3. Identify usable video and image files.
4. Extract video frames with timestamps.
5. Identify the corresponding experimental identifiers.
6. Extract measurable flame characteristics.
7. Prepare structured training features.
8. Establish ground-truth labels from documented experimental outcomes.
9. Train and evaluate AI models.
10. Store trained model weights, preprocessing configurations, experiment mappings, and evaluation results.

11. Expose the models through backend APIs.

The raw dataset must be preserved. Processed data must never overwrite original NASA files.

The platform must maintain traceability between model outputs and the underlying NASA experiment whenever a reliable relationship exists.

Do not assume that all videos, sensor records, and experimental table entries are automatically synchronized. Establish their relationships using actual metadata and documentation.

3. CORE AI MODEL ARCHITECTURE

FIREWISE-X should have five integrated intelligence components.

MODEL 1: AI Flame Detection and Segmentation Engine

Purpose:

Detect flame regions from experimental or uploaded videos and understand their visual characteristics.

Recommended models:

- YOLO segmentation or U-Net for flame segmentation.
- OpenCV for geometric feature extraction.
- Optical flow for motion analysis.

Input:

- NASA experimental videos.
- Researcher-uploaded videos.
- Compatible live video streams when supported.

Processing:

1. Read video frames.
2. Extract frames and timestamps.
3. Detect the flame region.
4. Segment the flame from its background.
5. Calculate flame area, perimeter, centroid, width, and height.
6. Track the flame across consecutive frames.
7. Estimate flame-front movement when spatial calibration is available.
8. Identify visible flame disappearance, with appropriate verification.

Output:

- Original video.
- AI segmentation overlay.
- Flame area timeline.
- Flame geometry.
- Motion analysis.

- Frame-level annotations.
- Analysis summary.

MODEL 2: Combustion Behavior Prediction Engine

Purpose:

Understand how experimental conditions influence combustion outcomes.

Input features may include:

- Fuel material.
- Fuel thickness.
- Airflow velocity.
- Oxygen concentration.
- Available pressure measurements.
- Relevant experimental environmental variables.

Recommended models:

- Random Forest baseline.
- XGBoost.
- LightGBM.
- Temporal models only when sufficient sequential data exists.

Possible prediction targets must be selected according to available documented ground truth:

- Flame extinction outcome.
- Combustion duration.
- Flame spread rate.
- Oxygen concentration changes.
- Other measured combustion outcomes.

Output:

- Predicted experimental outcome.
- Prediction uncertainty.
- Important influencing variables.
- Comparison against actual measurements when available.

MODEL 3: Temporal Flame Behavior Analysis

Purpose:

Understand how combustion changes over time instead of analyzing isolated images.

Input:

Sequential video frames and extracted flame features.

Processing:

- Flame growth analysis.

- Temporal feature extraction.
- Flame motion tracking.
- Changes in flame geometry.
- Observable extinction behavior.

Possible architecture:

- LSTM or temporal CNN.
- Temporal feature aggregation.
- Sequence-based anomaly detection when sufficient training data exists.

Output:

- Flame behavior timeline.
- Growth and decline trends.
- Temporal anomaly indicators.
- Observed versus predicted behavior.

MODEL 4: Multimodal Combustion Intelligence

Purpose:

Combine information from visual flame behavior and experimental measurements.

Inputs:

- Visual features from the flame analysis model.
- Experimental variables from PSI-25.
- Available synchronized sensor measurements.

Architecture:

- Visual feature encoder.
- Tabular feature encoder.
- Feature fusion layer.
- Prediction head.

Output:

- Combined combustion analysis.
- Supported target predictions.
- Model uncertainty.
- Feature contributions.

Only combine records when their experiment identities and temporal relationships can be established reliably.

MODEL 5: Explainable AI and Evidence Engine

Purpose:

Make AI outputs understandable and scientifically traceable.

Technologies:

- SHAP for tabular model explanations.
- Grad-CAM where appropriate for image models.
- Uncertainty estimation.
- Evidence retrieval from relevant NASA records and documents.

Output:

- Why the model produced a prediction.
- Which variables contributed most.
- What experimental evidence supports the analysis.
- How uncertain the prediction is.
- Which scientific limitations apply.

4. MAIN WEBSITE FUNCTIONALITIES

The website should have five main experiences.

MODULE 1: MISSION CONTROL DASHBOARD

This is the first page users see.

Purpose:

Provide an overview of the spacecraft combustion intelligence system.

Features:

- Mission overview.
- Platform status.
- AI model availability.
- Recent analyses.
- Recent anomaly assessments.
- Quick access to AI analysis.
- Quick access to pre-mission assessment.
- Scientific insight highlights.

Display only real backend statistics. Do not display fabricated live spacecraft telemetry.

Main actions:

- Launch AI Analysis.
- Start Pre-Mission Assessment.
- Open Astronaut Assistant.
- View Research Insights.

MODULE 2: FLAME ANALYSIS STUDIO

Purpose:

Allow researchers to analyze combustion videos.

User workflow:

1. Open Flame Analysis Studio.
2. Select a NASA experimental video or upload a compatible video.
3. Validate the video format.
4. Click Analyze Flame.
5. Backend extracts frames.
6. Flame detection and segmentation models process the frames.
7. Temporal analysis calculates flame characteristics.
8. Results are displayed visually.

Interface features:

- Original video viewer.
- AI segmentation viewer.
- Playback controls.
- Frame-by-frame inspection.
- Flame area timeline.
- Flame geometry charts.
- Flame movement analysis.
- Downloadable results.

The user should be able to compare the original and AI-processed video side by side.

MODULE 3: PRE-MISSION COMBUSTION ASSESSMENT

This should be one of the most important features of FIREWISE-X.

Purpose:

Help researchers and spacecraft safety engineers investigate possible combustion behavior under different experimental conditions before a space mission.

User workflow:

1. Select an available experimental fuel material.
2. Enter or select fuel thickness.
3. Specify oxygen concentration.
4. Specify airflow conditions.
5. Enter other supported environmental parameters.
6. Submit the assessment.
7. The backend validates the inputs.
8. The prediction model analyzes the conditions.
9. The system returns model-supported predictions.
10. The Evidence Engine explains the result.

Results should include:

- Predicted combustion outcome.
- Expected measurable behavior where supported.
- Important influencing variables.
- Prediction uncertainty.
- Relevant NASA experimental comparisons.
- Scientific limitations.
- Suggested further investigation.

Introduce a scenario comparison feature.

Researchers should be able to compare Scenario A, Scenario B, and Scenario C using different experimental parameters.

For example:

- Different oxygen concentrations.
- Different airflow conditions.
- Different fuel thicknesses.

The system should visualize how changing one or more parameters affects model predictions.

These are computational scenarios, not guaranteed physical outcomes.

MODULE 4: ASTRONAUT AI ASSISTANT

This is the signature interactive feature.

Purpose:

Provide an AI-assisted interface for astronauts and spacecraft safety personnel to interpret available combustion-related information.

The assistant should not simply be a generic chatbot.

It should be a domain-specific Microgravity Fire Safety Assistant.

Main capabilities:

A. Combustion Intelligence

- Explain flame behavior.
- Explain microgravity combustion concepts.
- Interpret available model outputs.
- Explain experimental conditions.

B. Visual Analysis Assistance

- Accept supported experimental video uploads.
- Summarize flame analysis results.
- Explain changes in flame geometry.
- Identify observed unusual visual patterns.

C. Sensor Interpretation

- Accept supported sensor measurements.
- Examine oxygen, pressure, airflow, and other available parameters.
- Identify deviations from an explicitly defined reference baseline.
- Explain potential implications.
- Distinguish sensor anomalies from confirmed fire events.

D. Safety Guidance

- Provide relevant, source-grounded explanations.
- Retrieve applicable spacecraft fire safety documentation.
- Explain the limitations of available information.
- Direct users to approved mission-specific emergency procedures.

E. Conversational Interface

Example questions:

- What does the current flame behavior indicate?
- Why is the flame area increasing?
- How has the oxygen measurement changed?
- Does this measurement deviate from the reference condition?
- Explain the latest AI analysis.
- What experimental conditions are most relevant to this result?

The assistant should provide:

- Concise answers.
- Supporting evidence.
- Relevant experimental references.
- Confidence and uncertainty where appropriate.
- Clear distinction between observations and AI inferences.

For an actual spacecraft deployment, the assistant must operate offline or on approved mission infrastructure, use validated models, and follow strict safety, cybersecurity, and human-approval requirements.

It must never independently activate fire suppression equipment or override crew instructions.

MODULE 5: RESEARCH INTELLIGENCE AND REPORTS

Purpose:

Provide a professional research workspace for scientific investigation.

Features:

- Searchable experimental evidence.
- Experiment comparisons.
- Flame analysis results.
- Model prediction history.
- Explainable AI results.

- Scenario comparison.
- Research visualizations.
- PDF report generation.

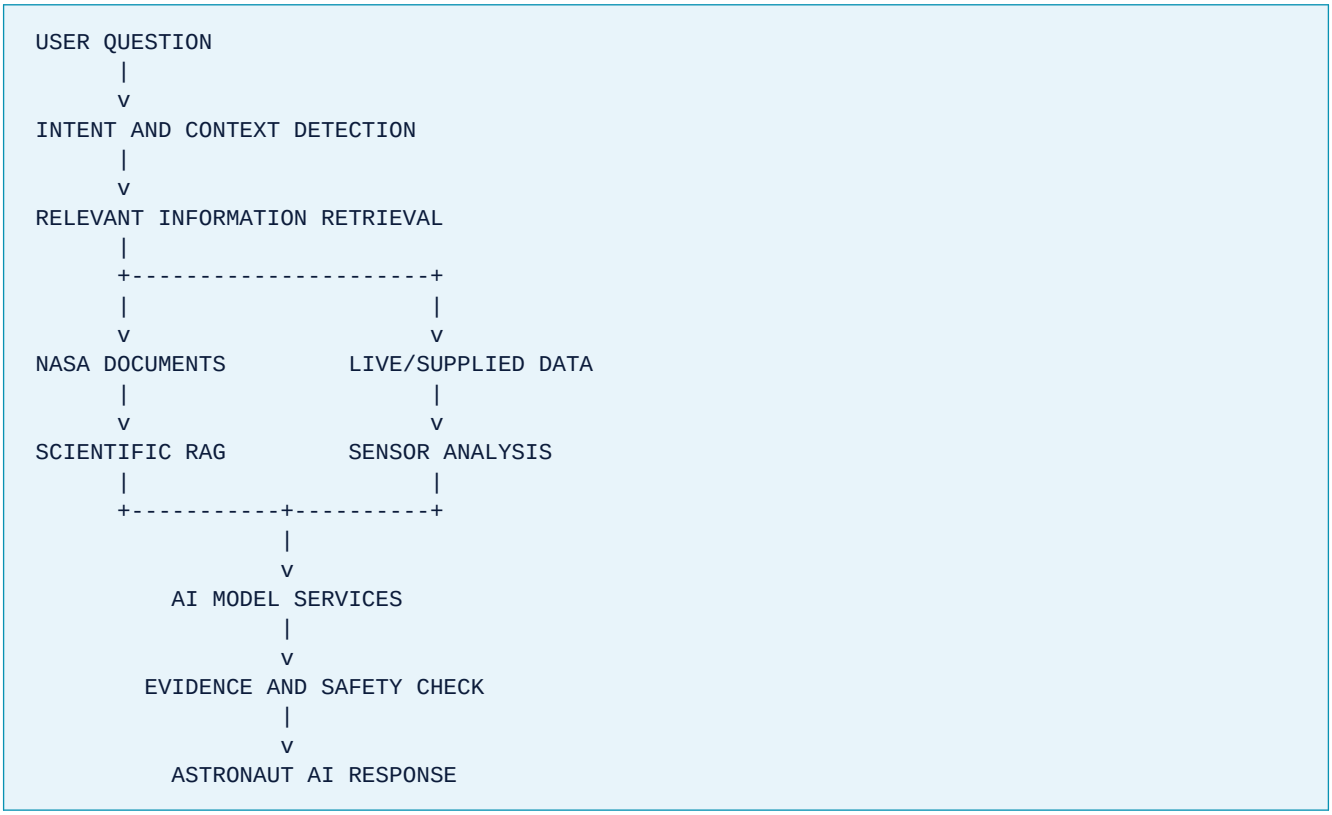
Report structure:

1. Experiment information.
2. Input conditions.
3. Video analysis.
4. Flame characteristics.
5. AI prediction.
6. Model explanation.
7. Experimental comparison.
8. Uncertainty and limitations.
9. References.

5. HOW THE ASTRONAUT AI ASSISTANT WORKS

The assistant should use a combination of language models, scientific document retrieval, structured sensor interpretation, and the trained combustion models.

Architecture:



The language model should not invent sensor measurements or combustion outcomes.

When the user asks about an actual experimental condition, the assistant should retrieve the relevant evidence or model output before answering.

When no relevant information is available, it should explicitly say so.

6. USER EXPERIENCE AND INTERFACE DESIGN

Design philosophy:

Simple, cinematic, scientific, and easy to use.

The platform should feel like a modern spacecraft research system, not a complicated engineering control panel.

Visual identity:

- Deep navy background.
- Electric cyan highlights.
- Soft blue interface elements.
- Orange flame indicators.
- Minimal glass-style cards.
- Clean typography.
- Smooth transitions.

Main navigation:

1. Overview.
2. Flame Studio.
3. Pre-Mission Assessment.
4. Astronaut AI.
5. Research Reports.

The NASA dataset should not be a primary navigation item. It should be managed internally by the backend.

Landing page:

Hero title:

"Understanding Fire Beyond Earth."

Subtitle:

"AI-powered microgravity combustion intelligence for safer space exploration."

Primary buttons:

- Explore AI Intelligence.
- Launch Pre-Mission Assessment.

Hero visual:

A cinematic microgravity flame inside a spacecraft laboratory.

Below the hero:

- What FIREWISE-X does.
- Flame Analysis.
- Pre-Mission Intelligence.
- Astronaut AI Assistant.
- Research Capabilities.

Use subtle animations:

- Floating flame visual.
- Smooth card transitions.
- Animated charts.
- Processing progress.
- Gentle background particles.

Avoid excessive neon effects, complicated menus, and distracting animations.

The platform must be responsive and accessible.

7. BACKEND TECHNOLOGY

Frontend:

- Next.js.
- TypeScript.
- Tailwind CSS.
- shadcn/ui.
- Framer Motion.
- Recharts.

Backend:

- Python.
- FastAPI.
- PostgreSQL.
- Object storage.

AI:

- PyTorch.
- OpenCV.
- YOLO segmentation or U-Net.
- XGBoost.
- SHAP.
- LSTM or temporal CNN where justified.
- RAG for scientific documents.
- An appropriately selected language model for the assistant.

Backend API groups:

```
/api/flame/analyze
/api/flame/results/{analysis_id}
/api/assessment/predict
/api/assessment/compare
/api/assistant/chat
/api/assistant/analyze-sensors
/api/research/reports
/api/research/experiments
```

All model inference must happen on the backend. The frontend should communicate through authenticated APIs.

8. DATA SECURITY AND SCIENTIFIC RELIABILITY

The system must:

- Preserve original NASA experimental files.
- Track data provenance.
- Validate incoming measurements.
- Maintain experiment identifiers.
- Avoid train-test leakage.
- Split training and evaluation data by independent experiments.
- Record model versions.
- Display uncertainty.
- Log predictions and analysis history.
- Distinguish observed values from inferred values.
- Provide evidence for scientific claims.

Never represent a model prediction as a verified experimental measurement.

Never label an unvalidated output as an actual spacecraft fire-risk level.

A production astronaut safety application would require independent validation, operational testing, approved hardware/software integration, and human oversight.

9. DEVELOPMENT ROADMAP

PHASE 1 — FUNCTIONAL WEBSITE FOUNDATION

- Build landing page.
- Build mission dashboard.
- Build navigation and responsive design.
- Build the five primary experiences.
- Connect frontend to backend.

PHASE 2 — NASA DATA INGESTION

- Import PSI-25.
- Audit files.
- Extract experimental metadata.
- Process usable videos.
- Establish video-to-experiment mappings.
- Prepare scientifically valid labels.

PHASE 3 — AI MODEL DEVELOPMENT

- Train flame segmentation model.
- Develop flame feature extraction.
- Develop tabular combustion prediction.
- Develop temporal analysis.
- Evaluate multimodal fusion feasibility.
- Implement explainability.

PHASE 4 — PRE-MISSION INTELLIGENCE

- Build assessment input interface.
- Connect prediction APIs.
- Implement scenario comparisons.
- Display uncertainty and experimental evidence.

PHASE 5 — ASTRONAUT AI ASSISTANT

- Implement scientific RAG.
- Connect combustion model APIs.
- Implement sensor interpretation.
- Implement source-grounded answers.
- Add safety and uncertainty checks.

PHASE 6 — FINAL INTEGRATION

- Research reports.
- Authentication and access control.
- Model monitoring.
- Testing.
- Deployment.

10. FINAL PRODUCT VISION

FIREWISE-X should become a scientific intelligence platform that transforms historical microgravity combustion experiments into useful AI-assisted insights for future space exploration.

NASA PSI-25 provides the experimental foundation.

Computer vision provides flame understanding.

Machine learning provides experimentally grounded predictions.

Multimodal analysis combines different sources of evidence.

Explainable AI makes the predictions understandable.

The Astronaut AI Assistant makes the scientific intelligence accessible through natural language.

The Pre-Mission Assessment helps researchers investigate conditions before launch.

The ultimate vision is:

> *"From NASA combustion experiments to AI-assisted spacecraft fire intelligence — helping researchers understand potential fire behavior before missions and supporting astronauts with evidence-based insights during spaceflight."*

FIREWISE-X: Understand the Flame. Anticipate the Risk. Protect the Mission.